



United International University (UIU)

Dept. of Computer Science & Engineering (CSE)

Exam Name: Mid Term Exam Trimester: Summer 2025

Course Code: PMG 4101, Course Title: Project Management

Total Marks: 30

Duration: 1 hour 30 minutes

Any examinee found adopting unfair means will be expelled from the trimester / program as per UIU disciplinary rules.

Answer all the questions.

Consider the following Scenario:

A leading multinational retail brand, which has seen rapid growth over the past few years, has approached your company to develop a comprehensive Smart Retail System to better integrate their in-store experience with an e-commerce platform. This system will enable them to streamline their inventory management, enhance customer engagement, and provide a seamless online and offline shopping experience. The client requires a solution that can track product stocks in real-time across all their stores, manage product availability, and offer personalized discounts based on customer preferences and shopping habits. Additionally, they need a system that can suggest products to customers based on their previous purchases and browsing history.

One of the primary features requested by the client is a mobile app that allows customers to browse the product catalog, check product availability in nearby locations, make purchases in-store, or choose home delivery. The backend system will be connected to this app to synchronize purchase information and product stocks. Additionally, the client has asked for an AI-powered system that can predict demand by monitoring sales trends, which would help optimize stocks and boost supply chain effectiveness. The project also requires third-party payment gateways for a smooth transaction process, ensuring that all functions across in-store and online purchases are fully aligned. However, the AI features, which are intended for future phases, are not yet prioritized, as the client is still in the early stages of exploring their possibilities. For now, the focus is on building the core functionalities of the inventory management system, customer engagement tools, and a user-friendly mobile application.

Due to the size and complexity of the project, the client has requested that it be broken down into smaller, manageable modules. The development of each module will be evaluated by the client before proceeding to the next. This approach allows flexibility to adapt the system as new features are requested and ensures that the system can evolve to meet the dynamic needs of the retail sector. To accommodate future scalability, the system must be designed to handle large amounts of data as the client plans to expand its operations into new regions. A key concern is the integration of the system with the existing software infrastructure, which could pose risks for data consistency and synchronization across different platforms. Additionally, considering the potential need for advanced data analytics and AI capabilities in the future, the team recognizes the importance of bringing in a data-science specialist to develop the AI/ML features as needed. Given the increasing security concerns with digital transactions, a cybersecurity expert will also be required to ensure the safety of customer data and to prevent potential breaches, particularly as the mobile app and e-commerce platform will handle sensitive payment information.

Realizing the size of the project, your team has decided to prioritize and deliver modules incrementally, with key milestones focused on inventory management, product catalog integration, and the development of the mobile app. After the completion of each module, regular meetings will be held with the client to evaluate if any new features or adjustments are needed to keep the system competitive in the fast-evolving retail market. As a project manager, you will oversee the entire process, ensuring smooth collaboration between teams, managing resources effectively, and keeping the project within scope, budget, and timeline.

According to this scenario, answer the following questions.

- 1.** Choose a Software Engineering **Methodology** for developing the above-mentioned project scenario and discuss the rationale for choosing that methodology. Discuss the drawbacks of *Xtreme Programming (XP)* and explain which type of project this methodology will be useful.

[CO1] 5
- 2.** Write down the **Vision** and **Scope** Document for the project described above following proper outline.

[CO2] 5
- 3.** As a project manager if you want to generate a proper Work Breakdown Structure (WBS), state the rules that you would suggest to your team. Develop the WBS for the scenario described above and estimate (duration unit of month) for **two** major **WBS pieces**/components for the above scenario following the principles and rules.

[CO3] 5
- 4.** Write down activities of the *moderator* during the '*Kick off Meeting*' for *Delphi Wideband Estimation (DWE)* technique. Suppose you want to form an estimation team consisting of **four members**. Show the steps of estimation session (**one** individual estimation form and tentative convergence chart of duration) for the above-described project, according to DWE method.

[CO3] 5
- 5.** Now as a project manager you are preparing for Risk Mitigation. Develop a *Risk Plan Sheet* showing at least **three** major risks for the above scenario.

[CO2] 5
- 6.** Progress report submission and presentation on Work Breakdown Structure and Delphi Wideband Estimation Technique for a group project have been evaluated earlier (**Do not answer this question, already completed in classroom**).

[CO3] 5